

Solution Requirements

Date	29 SEPTEMBER 2026
Team ID	SWTID-2026-6684
Project Name	ComicCraft-AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Currently stateless/single-user session without mandatory sign-in; allows immediate guest creation	Low
2	Authorization levels	Single public user role for creating, previewing, and downloading comics	Medium

3	External interfaces	[Must interface with Google Gemini Generative AI API (gemini-1.5-flash, gemini-1.5-pro) and Hugging Face Diffusers (stable-diffusion-v1-5	High
4	Transactions processing	Synchronous execution pipeline: accepts prompt $\rightarrow$ generates outline $\rightarrow$ generates story $\rightarrow$ renders images $\rightarrow$ constructs layout $\rightarrow$ compiles PDF	High
5	Reporting	Generates dynamic structured panel previews on web interface and produces downloadable PDF files formatted with FPDF	High
6	Business rules, etc.	Restricts outlines strictly to 5 structured JSON panels containing panel number, title, scene description, and image prompt.	High
7	Compliance to laws or regulations	Adheres to safe content practices via prompt sanitization and reliance on model safety filters	Medium

8	<i>Other</i>	Provides both browser-based HTML form routes (/generate) and direct programmatic JSON endpoints (/generate-comic/json)	High
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Efficient generation of text outlines and processing of images	Text outlines in < 3s; full comic generation & rendering completed smoothly.
2	Scalability	Asynchronous FastAPI request handling capability.	Capable of serving concurrent API client and web user sessions
3	Security & Data Privacy	Secure handling of API keys via environment variables (.env)	Zero hardcoded credentials in source code; sanitized image filenames
4	Reliability & Availability	Robust error handling for AI responses and JSON parsing errors.	Graceful fallback error messages via FastAPI <code>HTTPException</code> without application crashes.

5	Usability & Accessibility	<i>[Describe user experience standards and accessibility compliance]</i>	Mobile-friendly layout using Jinja2 templates, standard readable fonts (DejaVu) in PDF.
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